

COURIER MANAGEMENT SYSTEM (CMS)

User Story

The Courier Management System (CMS) is designed to simplify and manage the complete courier delivery process through a centralized system. The system allows customers to register, log in, create courier bookings, enter sender and receiver details, and track the delivery status of their parcels.

Customer User Stories

1. Registration and Login

As a customer, I want to register and log in to the system so that I can securely access courier services.

2. Courier Booking

As a customer, I want to book a courier by entering sender, receiver, parcel, and delivery details so that my parcel can be shipped.

3. Courier Tracking

As a customer, I want to track my courier using a tracking ID so that I can know the current status and location of my parcel.

4. Delivery Status

As a customer, I want to view the delivery status such as Booked, Picked Up, In Transit, Out for Delivery, and Delivered so that I can monitor my shipment.

5. Booking History

As a customer, I want to view my previous courier bookings so that I can check my shipment history.

6. Payment

As a customer, I want to make payment for courier services so that my booking can be confirmed.

Staff/Admin User Stories

7. Manage Customers

As an admin, I want to manage customer information so that user details can be maintained securely.

8. Manage Courier Bookings

As a staff member, I want to view and manage courier bookings so that parcels can be processed efficiently.

9. Update Courier Status

As a staff member, I want to update the courier status at each delivery stage so that customers receive accurate tracking information.

10. Manage Delivery

As a delivery staff member, I want to view assigned parcels and update their delivery status so that parcels can be delivered successfully.

11. Generate Reports

As an admin, I want to generate reports about bookings, deliveries, payments, and customers so that I can monitor the overall courier operations.

Expected Outcome

The Courier Management System provides a simple and secure platform for managing courier bookings, tracking parcels, handling deliveries, maintaining customer information, and generating reports. It reduces manual work, improves delivery tracking, and provides customers with up-to-date information about their shipments.

1. INTRODUCTION

1.1 Project Overview:

The Courier Management System (CMS) is a web-based application developed to simplify and manage the complete process of courier services. The system provides a centralized platform for customers, staff, delivery personnel, and administrators to manage courier bookings, parcel information, delivery status, payments, and tracking details efficiently.

In the traditional courier management process, maintaining customer details, booking information, parcel records, and delivery updates manually can be time-consuming and may lead to errors. The proposed Courier Management System reduces these difficulties by storing all information digitally and providing an organized method for managing courier operations.

The system allows customers to create an account and securely log in to access courier services. Customers can book a courier by providing sender details, receiver details, parcel information, pickup location, delivery location, and other required information. After booking, the system generates a unique tracking ID that can be used to monitor the parcel throughout the delivery process.

The Courier Management System also provides a tracking facility through which customers can view the current status of their parcels. The delivery status can be updated at different stages, such as Booked, Picked Up, In Transit, Out for Delivery, and Delivered. This helps customers receive accurate and timely information about their shipments.

Administrators and staff members can manage customer records, courier bookings, parcel details, delivery assignments, payments, and status updates through the system. The administrator can also generate reports related to courier

bookings and deliveries, which helps in monitoring the overall operation of the courier service.

The main objective of the Courier Management System is to provide a simple, secure, reliable, and user-friendly platform for managing courier activities. By replacing manual processes with a digital system, the project helps reduce paperwork, improve data accuracy, save time, and make courier tracking more convenient for customers and staff.

Overall, the Courier Management System provides an efficient solution for managing the complete courier lifecycle, from booking and pickup to transportation, tracking, and final delivery.

1.2 PURPOSE

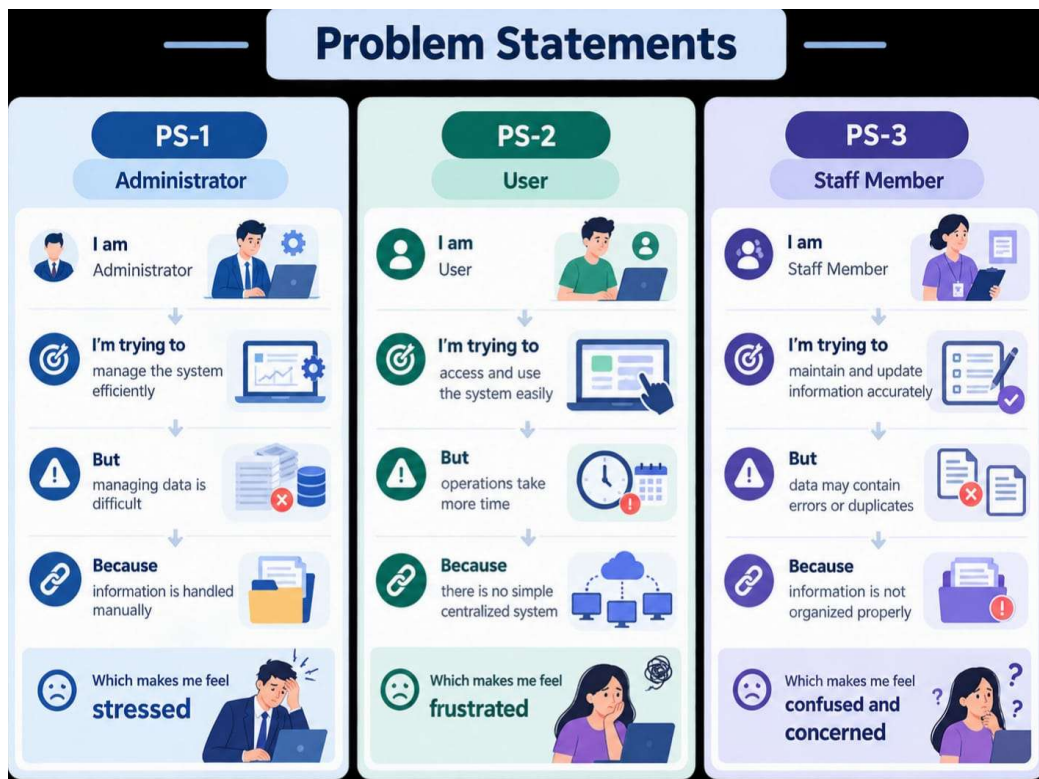
The main purpose of the **Courier Management System (CMS)** is to develop a simple and efficient digital platform for managing courier and delivery activities. The system is designed to reduce manual work and make the process of courier booking, tracking, delivery, and record management easier and more organized.

The system allows customers to register, log in, book courier services, enter sender and receiver details, and track their parcels using a unique tracking ID. It also enables staff members to manage courier bookings, update parcel status, assign deliveries, and maintain customer and shipment information.

The system provides real-time status updates such as **Booked, Picked Up, In Transit, Out for Delivery, and Delivered**, helping customers know the progress of their shipments. Administrators can manage users, bookings, payments, delivery information, and generate reports for better system management.

2. IDEATION PHASE

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Administrator	manage the system efficiently	managing data is difficult	information is handled manually	stressed
PS-2	User	access and use the system easily	operations take more time	there is no simple centralized system	frustrated
PS-3	Staff Member	maintain and update information accurately	data may contain errors or duplicates	information is not organized properly	confused and concerned



2.2 Empathy Map Canvas

User Persona: Administrator

- Think & Feel: Concerned about data management and system efficiency
- Hear: User complaints and operational difficulties
- See: Records, system data, and user activities
- Say & Do: Manage information, monitor activities, and perform regular follow-ups



2.3 Brainstorming & Idea Prioritization

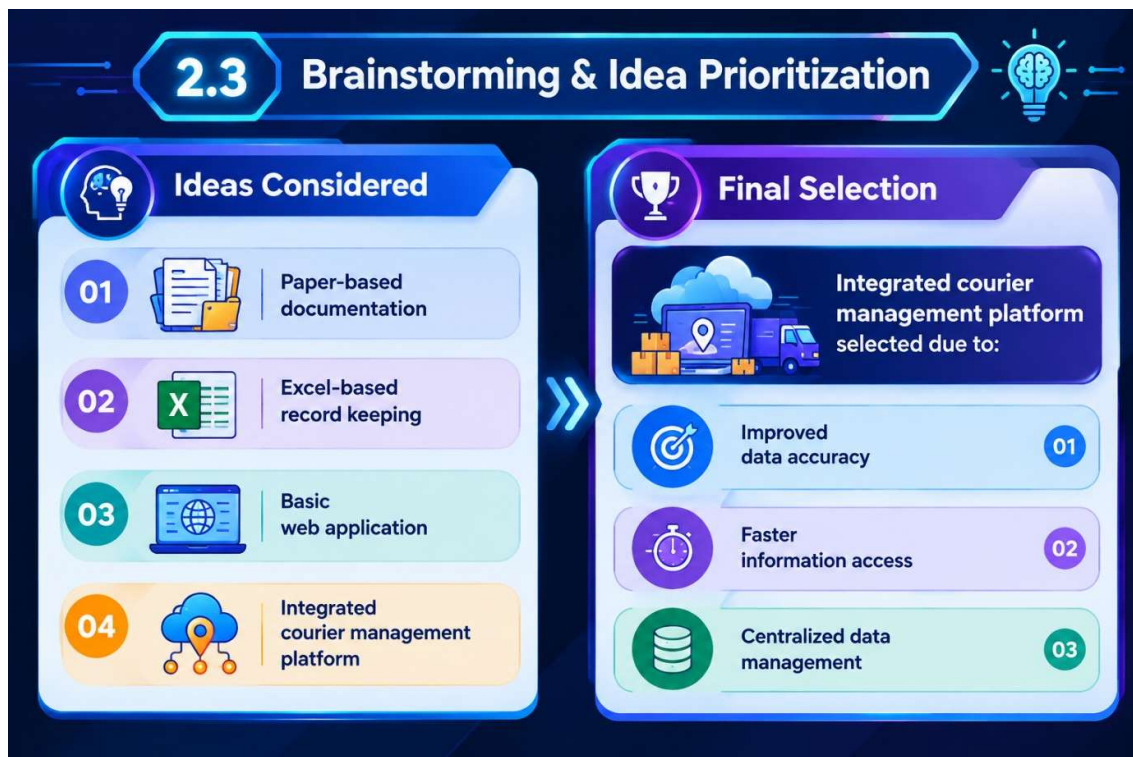
Ideas Considered

- Paper-based documentation
- Excel-based record keeping
- Basic web application
- Integrated courier management platform

Final Selection

Integrated courier management platform selected due to:

- Improved data accuracy
- Faster information access
- Centralized data management



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Customer Inquiry → Courier Booking → Shipment Creation → Agent Assignment → Shipment Transit → Delivery Update → Invoice Generation → Delivery Completion → Reporting

Customer Journey:

- **Customer Inquiry** – Customer provides the required courier details.
- **Courier Booking** – Shipment details are registered in the system.
- **Shipment Creation** – A shipment record and tracking information are generated.
- **Agent Assignment** – A suitable delivery agent is assigned to the shipment.
- **Shipment Transit** – The shipment moves from the source location to the destination.
- **Delivery Update** – Shipment status is updated during the delivery process.
- **Invoice Generation** – Charges and invoice details are recorded.
- **Delivery Completion** – The shipment is marked as successfully delivered.
- **Reporting** – Shipment and delivery information is used for reports and monitoring.



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User registration and login
FR-2	Courier pickup and delivery scheduling
FR-3	Tracking number generation
FR-4	Shipment status management
FR-5	Customer notification management
FR-6	Courier history and transaction records

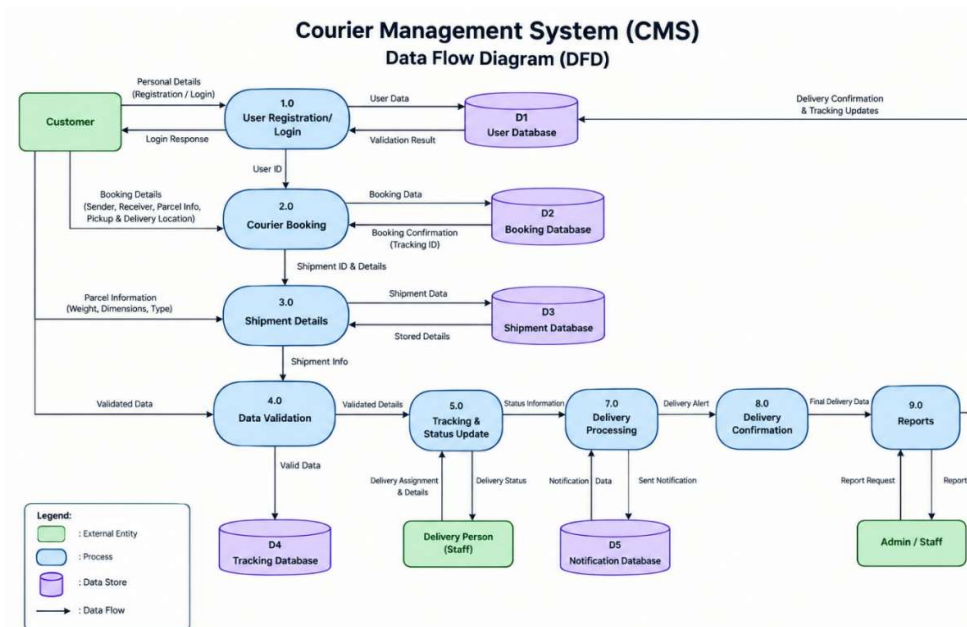
Non-Functional Requirements

NFR	Description
Maintainability	Easy to maintain and update the system
Data Integrity	Ensures accurate and consistent courier information
Compatibility	Works properly across commonly used devices and browsers
Backup & Recovery	Protects important data through regular backup and recovery mechanisms
Accessibility	Provides easy access and navigation for different users
Response Time	Provides quick response while performing system operations

3.3 Data Flow Diagram

Flow:

Customer Input → User Registration/Login → Courier Booking → Shipment Details → Data Validation → Tracking & Status Update → Delivery Processing → Notification → Delivery Confirmation → Reports



4. FUNCTIONAL REQUIREMENTS

The Courier Management System (CMS) is designed to manage courier booking, shipment tracking, customer information, delivery details, and courier staff activities in an organized manner. The system provides a centralized platform where customers, courier staff, and administrators can manage courier-related activities efficiently.

The functional requirements describe the operations that the system must perform. These requirements define how users interact with the system and how the system processes courier information.

4.1 User Registration

The system shall allow new customers to register by providing necessary details such as:

- Customer Name
- Email Address
- Mobile Number
- Address
- City
- State
- Password

The system validates the entered information before creating the customer account. Each registered customer can log in using their credentials.

4.2 User Login

The system shall provide a secure login facility for registered users. Users enter their username/email and password to access the system.

The system verifies the credentials and provides access according to the user's role.

Different users may have different access permissions:

- Administrator
- Courier Staff
- Customer

4.3 Customer Management

The system shall maintain complete customer information. The administrator can view and manage customer records.

Customer information includes:

- Customer ID
- Customer Name
- Contact Number
- Email
- Address
- City
- State
- Pincode

The system shall allow authorized users to add, view, update, and manage customer information.

4.4 Courier Booking

The system shall allow customers or authorized staff to create a new courier booking.

The booking form may contain:

- Sender Name
- Sender Address
- Sender Phone Number
- Receiver Name
- Receiver Address
- Receiver Phone Number
- Courier Type
- Package Description
- Package Weight
- Booking Date
- Delivery Type
- Payment Details

After successful booking, the system generates a unique courier or tracking number.

4.5 Courier Details Management

Each courier shall have a unique record in the system. The system stores all important information related to the courier.

The courier record includes:

- Courier ID
- Tracking Number
- Sender Details

- Receiver Details
- Package Details
- Weight
- Booking Date
- Expected Delivery Date
- Courier Status
- Delivery Staff
- Payment Status

Authorized users can view and update courier details.

5. COURIER TRACKING

Courier tracking is one of the main functions of the system. Customers can enter their tracking number to check the current status of their courier.

Possible courier statuses include:

- Booked
- Picked Up
- In Transit
- At Distribution Center
- Out for Delivery
- Delivered
- Delivery Attempted
- Cancelled

The system displays the latest available courier status to the customer.

5.1 Tracking Number

A unique tracking number is generated for every courier booking. The tracking number is used to identify and track the shipment throughout the delivery process.

5.2 Tracking History

The system can maintain the movement history of the courier.

The history may contain:

Date	Location	Status
25-09-2026	Chennai	Booked
26-09-2026	Chennai Hub	Picked Up
27-09-2026	Trichy Hub	In Transit
28-09-2026	Destination Hub	Out for Delivery
28-09-2026	Customer Address	Delivered

6. DELIVERY MANAGEMENT

The system shall help courier staff manage deliveries efficiently.

The administrator can assign courier shipments to delivery staff. The assigned staff member can view the courier information and update the delivery status.

Delivery information includes:

- Courier ID
- Tracking Number
- Delivery Staff
- Delivery Address
- Receiver Name
- Contact Number
- Delivery Date
- Delivery Status

When the courier is successfully delivered, the staff can update the status as Delivered.

6.1 Delivery Assignment

The administrator can assign a courier to a particular delivery employee.

The system stores:

- Employee ID
- Employee Name
- Courier ID
- Assignment Date
- Delivery Address
- Delivery Status

6.2 Delivery Confirmation

After delivering the courier, the delivery staff can update the delivery status. The system records the delivery date and time.

The system may also store confirmation information such as receiver name or delivery confirmation.

7. PAYMENT MANAGEMENT

The system shall maintain payment information related to courier bookings.

Payment details may include:

- Payment ID
- Courier ID

- Customer ID
- Amount
- Payment Method
- Payment Date
- Payment Status

Payment status may be:

- Pending
- Paid
- Failed
- Refunded

The administrator can view payment information and generate payment-related reports.

8. COURIER SEARCH

The system shall provide a search facility for administrators and authorized users.

Users can search courier information using:

- Tracking Number
- Courier ID
- Customer Name
- Receiver Name
- Mobile Number
- Courier Status
- Booking Date

The search function reduces the time required to locate courier records.

9. ADMINISTRATOR MANAGEMENT

The administrator has complete control over the Courier Management System.

The administrator can:

- Manage customers
- Manage courier bookings
- Manage courier records
- Manage employees
- Assign deliveries
- Update courier status
- Manage payment information
- View reports
- Monitor system activities

The administrator can also maintain accurate and updated information in the system.

10. REPORTS AND DASHBOARD

The system shall provide reports and a dashboard for administrators.

The dashboard can display:

- Total Couriers
- Total Customers
- Delivered Couriers
- Pending Couriers
- In-Transit Couriers
- Cancelled Couriers
- Total Payments

Reports can be generated based on courier status, date, customer, and delivery information.

The reports help administrators understand the overall performance of the courier service.

11. HARDWARE REQUIREMENTS

Hardware requirements describe the physical equipment needed to operate the Courier Management System.

11.1 Server Requirements

The server should have sufficient processing power and storage capacity to store application and database information.

Recommended server configuration:

Component	Requirement
Processor	Intel Core i5 or equivalent
RAM	8 GB or above
Hard Disk	256 GB SSD or above
Network	LAN / Internet Connection
Monitor	15-inch or above
Keyboard	Standard Keyboard
Mouse	Standard Mouse

11.2 Client Computer Requirements

The customer, administrator, and courier staff can access the system using a computer or laptop.

Recommended configuration:

Component	Requirement
Processor	Intel Core i3 or above
RAM	4 GB or above
Storage	128 GB or above
Display	1366 × 768 or above
Network	Internet Connection
Browser	Modern Web Browser

11.3 Network Requirements

An internet or local network connection is required for communication between the users and the application server.

A stable network connection helps users:

- Login to the system
- Book couriers
- Track shipments
- Update delivery information
- Access reports
- Communicate with the database

12. SOFTWARE REQUIREMENTS

The software requirements describe the software technologies required to develop and operate the Courier Management System.

12.1 Operating System

The system can be developed and deployed using:

- Windows 10/11
- Linux
- macOS

For development, Windows or Linux can be used depending on the development environment.

12.2 Front-End Technologies

The front end provides the user interface of the system.

The following technologies can be used:

HTML:

HTML is used to create the structure of web pages.

CSS:

CSS is used for designing web pages, layouts, colors, fonts, forms, tables, and responsive interfaces.

JavaScript:

JavaScript is used to provide interactive functionality and client-side validation.

12.3 Back-End Technology

The backend handles application logic, database communication, authentication, and data processing.

A suitable backend technology can be used to implement:

- User authentication
- Courier booking
- Customer management
- Tracking
- Delivery management
- Payment management
- Report generation

12.4 Database

A relational database is required to store system information.

The database can contain tables such as:

- Customer
- Courier
- Employee
- Delivery
- Payment
- Tracking
- User

The database maintains relationships between these entities and helps retrieve information efficiently.

12.5 Web Browser

The system can be accessed using modern web browsers such as:

- Google Chrome

- Microsoft Edge
- Mozilla Firefox

The browser allows users to access the web-based Courier Management System.

13. DETAILED FUNCTIONAL REQUIREMENTS

13.1 Customer Module

The customer module provides facilities for customers to interact with the courier service.

The customer can:

1. Register an account.
2. Login to the system.
3. Update profile information.
4. Book a courier.
5. Enter sender and receiver details.
6. View courier information.
7. Track courier status.
8. View booking history.
9. View payment details.
10. Check delivery status.

13.2 Courier Module

The courier module manages all shipment-related information.

The system shall:

1. Generate a unique courier ID.
2. Generate a tracking number.
3. Store sender information.
4. Store receiver information.
5. Store package details.
6. Store package weight.
7. Calculate or store delivery charges.
8. Maintain courier status.
9. Maintain booking date.
10. Maintain expected delivery date.

13.3 Tracking Module

The tracking module allows users to monitor courier movement.

The system shall allow users to enter the tracking number and retrieve courier information.

The system displays:

- Tracking Number
- Current Status
- Current Location
- Booking Date
- Expected Delivery Date
- Delivery Status

The tracking information is updated by authorized staff.

13.4 Employee Module

The employee module manages courier delivery employees.

The administrator can:

- Add employees
- Update employee information
- View employee details
- Assign couriers
- Monitor delivery status

Employee information may include:

- Employee ID
- Employee Name
- Phone Number
- Email
- Address
- Role
- Status

13.5 Delivery Module

The delivery module manages the final delivery process.

The system shall allow authorized staff to:

1. View assigned couriers.
2. View receiver details.
3. View delivery address.
4. Update delivery status.

5. Record delivery date.
6. Confirm successful delivery.

The delivery status can be changed according to the actual delivery progress.

13.6 Payment Module

The payment module maintains payment information.

The system records:

- Payment ID
- Customer ID
- Courier ID
- Amount
- Payment Method
- Payment Date
- Payment Status

The system can display whether the payment is completed or pending.

13.7 Notification Module

The system may provide notifications whenever an important courier event occurs.

Notifications can be provided for:

- Successful booking
- Courier pickup
- Courier dispatched
- Courier reaching destination hub
- Out for delivery
- Successful delivery
- Delivery failure

Notifications help customers stay informed about their shipments.

13.8 Validation Requirements

The system shall validate data entered by users.

Examples include:

- Email must be in valid format.
- Mobile number must contain a valid number of digits.
- Required fields cannot be empty.
- Package weight must be a valid numeric value.
- Tracking number must be unique.

- Password fields must follow the required format.
- Duplicate customer records should be avoided where applicable.

Validation helps maintain accurate and reliable data.

14. SECURITY REQUIREMENTS

Security is an important requirement of the Courier Management System.

The system shall provide:

User Authentication

Only registered users should be able to access protected system functions.

Role-Based Access

Different users should have different permissions.

For example:

Administrator: Full system access.

Courier Staff: Access to assigned courier and delivery information.

Customer: Access to their own profile, bookings, payments, and tracking information.

Data Protection

Customer and courier information should be protected from unauthorized access.

Password Security

Passwords should not be stored in plain text. Secure password handling should be used in the application.

15. PERFORMANCE REQUIREMENTS

The system should respond quickly to user requests.

The application should:

- Load pages efficiently.
- Process courier searches quickly.
- Retrieve tracking information without unnecessary delay.
- Handle multiple users.
- Maintain database consistency.
- Generate reports efficiently.

The system should continue to function reliably when the number of courier records increases.

16. RELIABILITY REQUIREMENTS

The system should provide reliable courier information.

Important requirements include:

- Accurate database records.
- Proper validation.
- Prevention of duplicate records.
- Regular database backup.
- Error handling.
- Recovery from unexpected failures.

The system should minimize data loss and maintain the consistency of courier records.

17. USABILITY REQUIREMENTS

The system interface should be simple and easy to understand.

The user interface should provide:

- Clear navigation menus.
- Simple forms.
- Readable text.
- Proper buttons and labels.
- Search functionality.
- Responsive design.
- Clear error messages.
- Confirmation messages after successful operations.

The system should be usable by customers and employees with basic computer knowledge.

18. PROJECT DESIGN PHASE

18.1 Problem–Solution Fit

The **Courier Management System (CMS)** addresses the difficulties caused by manual courier operations and unorganized data management. The system provides a centralized platform for managing customer details, courier bookings, shipment information, delivery status, and related records.

By replacing manual processes with a digital system, CMS helps reduce data entry errors, improve information accessibility, save time, and provide better coordination between users and administrators. This solution supports a more organized and efficient courier management process.



18.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual courier management and unorganized shipment records
Solution	Salesforce Courier Management System
Innovation	Digital booking, tracking and automated status management
Impact	Faster operations and improved information accessibility
Business Model	Courier and logistics services
Scalability	Supports increasing customers, shipments and delivery operations

18.3 Solution Architecture

Components:

- Customer Management
- User Management
- Courier Booking

- Shipment Management
- Delivery Agent Management
- Tracking & Status Management
- Notification Management
- Invoice Management
- Reports & Dashboard



19. PROJECT PLANNING & SCHEDULING

19.1 Project Planning

- Agile methodology
- Project requirements → Design → Development → Testing
- Task-based execution
- Regular progress monitoring

- Milestone-based development
- Time and resource estimation used for project planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Velocity calculation used for estimation: Each Sprint = 6 Days = 20 Story Points

1. Product Backlog & Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Member1
Sprint-1	Developer Setup	USN-9	As an admin, I want user login, profiles and permission sets so that each CMS user (manager, agent, staff) gets appropriate access.	5	High	Member2
Sprint-1	Data Modelling	USN-10	As an admin, I want to register Branch and Staff records so that courier operations are organized by branch.	5	High	Member3
Sprint-1	Developer Setup	USN-11	As an admin, I want a basic app layout and navigation menu so that users can easily move between CMS modules.	2	Medium	Member4
Sprint-2	Data Modelling	USN-2	As an admin, I want to create Customer, Shipment, Delivery Agent, Branch, and Invoice objects with proper relationships so that courier data is	5	High	Member1

			stored and managed efficiently.			
Sprint-2	Automation	USN-12	As a staff member, I want the shipment status to update automatically to 'In Transit' and 'Delivered' so that manual status updates are avoided.	5	High	Member2
Sprint-2	Automation	USN-13	As a customer, I want automatic Email/SMS notifications at every shipment milestone so that I always know my parcel status.	4	High	Member3
Sprint-2	Data Modelling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	3	Medium	Member4
Sprint-2	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member1
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member2
Sprint-3	Automation	USN-14	As a staff member, I want invoices to be generated automatically when a shipment is delivered so that billing is accurate and instant.	5	High	Member3

Sprint-3	Security	USN-15	As an administrator, I want organization-wide defaults and sharing rules so that shipment and customer data is visible only to the right branch users.	5	High	Member4
Sprint-3	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member1
Sprint-3	Security	USN-16	As a manager, I want field history tracking on shipment status and payment fields so that every change is audited.	2	Medium	Member2
Sprint-4	Reports	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member3
Sprint-4	Reports	USN-17	As a branch manager, I want a monthly revenue and pending-payment report so that billing follow-ups are easy.	4	High	Member4
Sprint-4	Reports	USN-18	As a branch manager, I want an agent-wise delivery performance report so that I can reward top performers.	4	Medium	Member1
Sprint-4	Reports	USN-19	As a dispatcher, I want a delayed/undelivered shipment report so that pending	4	High	Member2

			deliveries can be escalated on time.			
Sprint-4	Reports	USN-20	As a manager, I want reports exportable to Excel/PDF so that they can be shared in meetings.	4	Low	Member3
Sprint-5	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member4
Sprint-5	Dashboards	USN-21	As a delivery agent, I want a mobile-friendly dashboard of today's assigned deliveries so that I can update statuses on the go.	4	Medium	Member1
Sprint-5	Dashboards	USN-22	As a manager, I want a revenue trend dashboard with charts so that business growth is visible at a glance.	4	Medium	Member2
Sprint-5	Customer Service	USN-23	As a customer, I want a public tracking page using a tracking number so that I can check my shipment status anytime.	5	High	Member3
Sprint-5	Customer Service	USN-24	As a customer, I want a feedback/rating option after delivery so that service quality can be measured.	3	Low	Member4

2. Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022
Sprint-5	20	6 Days	21 Nov 2022	26 Nov 2022	20	26 Nov 2022

Velocity = 20 Story Points per Sprint (constant). Total Project Scope = 100 Story Points across 5 Sprints. Burndown: 100 → 80 → 60 → 40 → 20 → 0 story points remaining at the end of each sprint.

19.2 Project Development Phase:

1. Real-time Salesforce project implementation for a Courier Management System (CMS).
2. Designing custom objects — Customer, Shipment, Delivery Agent, Branch, and Invoice — with master-detail and lookup relationships.
3. Creating page layouts & record types for different courier modules (booking, dispatch, delivery).
4. Writing validation rules to prevent incorrect shipment details (delivery date, address, mandatory fields).
5. Building automation with flows for shipment status updates, agent assignment, and invoice generation.
6. Developing reports & dashboards to track delivery performance, shipment status, and revenue.
7. Setting up security with profiles, permission sets, and sharing rules to protect sensitive shipment and billing data.
8. End-to-end application testing — from shipment booking to delivery confirmation and customer tracking.

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like courier booking, shipment tracking, and delivery management—can be built on the platform? You're in the right place.

Welcome to the Courier Management System project!

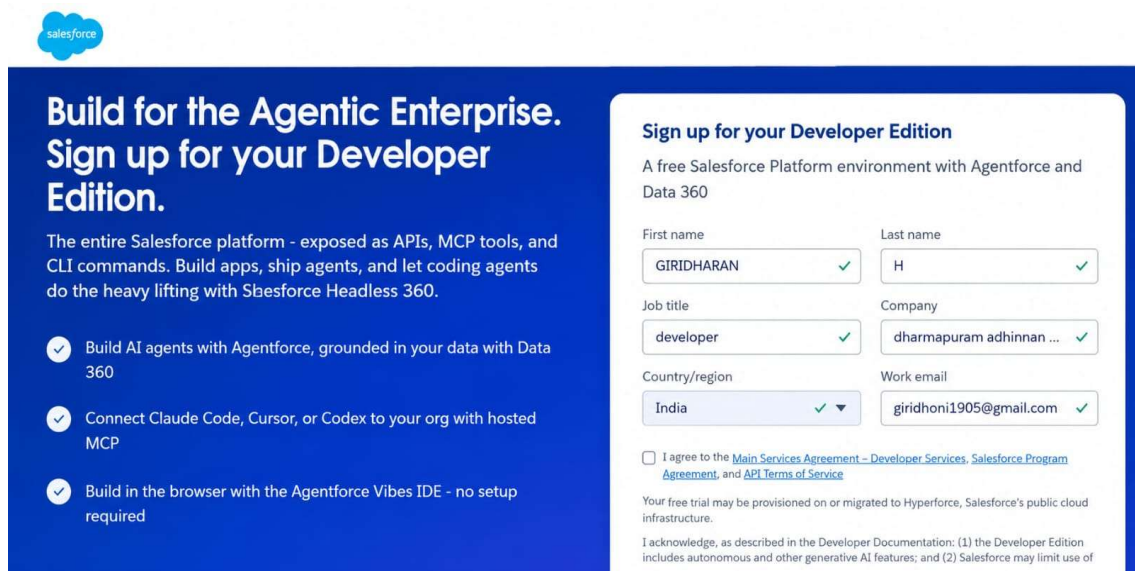
Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline courier workflows—from shipment booking and dispatch to agent assignment, delivery confirmation, and invoice generation.

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Creating Developer Account

1. Creating a developer org in Salesforce.
2. Go to <https://developer.salesforce.com/signup>
3. On the sign up form, enter the following details:
4. First name & Last name
5. Email



Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name: GIRIDHARAN ✓ Last name: H ✓

Job title: developer ✓ Company: dharmapuram adhinna... ✓

Country/region: India ✓ Work email: giridhoni1905@gmail.com ✓

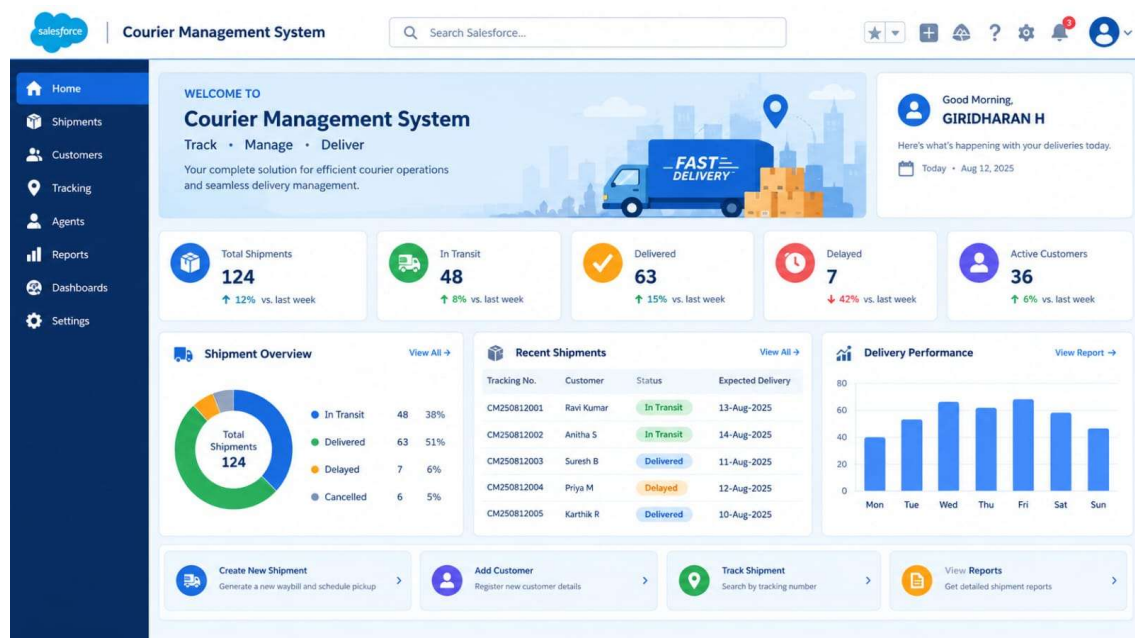
☐ I agree to the [Main Services Agreement - Developer Services, Salesforce Program Agreement, and API Terms of Service](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of

- 1) First name & last name
- 2) Email
- 3) Job title: developer
- 4) Company: Dharmapuram Adhinam Arts college
- 5) Country: India

Salesforce setup home page:



Milestone 2- object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

1. **Customer__c** — customer personal & contact info
2. **Shipment__c** — parcel details, source, destination, tracking
3. **Delivery_Agent__c** — agent profiles & branch details
4. **Branch__c** — branch offices
5. **Delivery_Update__c** — status updates per shipment

6. Invoice__c — billing for delivered shipments

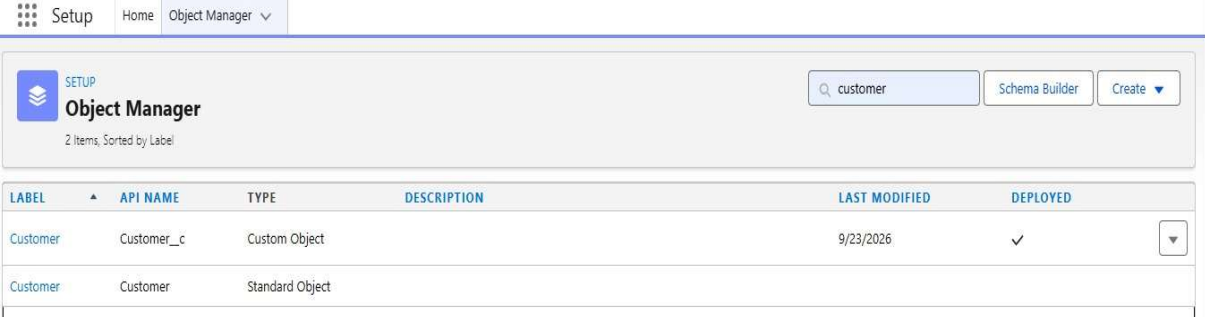
Relationships:

1. **Customer** → Shipment (Lookup)
2. **Shipment** → Delivery_Agent (Lookup)
3. **Delivery_Agent** → Branch (Lookup)
4. **Delivery_Update** → Shipment (Master-Detail)
5. **Invoice** → Shipment (Lookup)

Steps to implement:

Activity 1: create customer object

1. click the setup — object manager — create — customer object



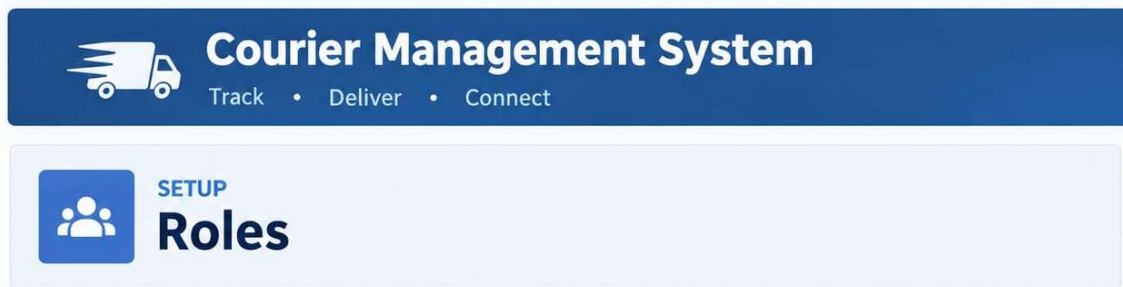
The screenshot shows the Salesforce Object Manager interface. At the top, there are tabs for 'Setup', 'Home', and 'Object Manager'. Below the tabs, there is a search bar with 'customer' entered, and buttons for 'Schema Builder' and 'Create'. The main content area displays a table with two rows of object information.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Customer	Customer__c	Custom Object		9/23/2026	✓
Customer	Customer	Standard Object			

2. Enter the following details:

- Label: customer
- Plural label: customers
- Record name: customer name (text)

ROLES:



Understanding Roles

Set up your Role Hierarchy to control how your organization handles courier operations and accesses data.

Sample Role Hierarchy

View other sample Role Hierarchies:



- View and edit data, manage couriers, branches and delivery operations for all users below.
- Can't access data of other admins.
- View and edit data, manage shipment and generate report for all users directly below.
- Can't access data of users above or at same level.

Workflows:

Courier Management System

Search Setup...

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👤

SETUP

Workflows

New Workflow

Flow Trigger Explorer

Workflow Definitions

All Workflows

50+ items • Sorted by Workflow Label • Filtered by All workflow definitions • Updated a minute ago

⚙️

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🔍

Workflow Label ↑	Process Type ▾	Active ▾	Template ▾	Package State ▾	Last Modified By ▾	Last Modified Date ▾
Add Shipment	Autolaunched Flow	☒	☐	Created	Admin	12-Aug-2025, 10:24 AM ▾
Update Shipment Status	Record-Triggered Flow	☒	☐	In Transit	Admin	11-Aug-2025, 04:18 PM ▾
Generate Delivery Notification	Screen Flow	☒	☒	Out for Delivery	Admin	10-Aug-2025, 02:37 PM ▾
Schedule Pickup Appointment	Scheduled Flow	☒	☐	Picked Up	Admin	09-Aug-2025, 11:12 AM ▾
Handle Return Request	Screen Flow	☒	☐	Returned	Admin	08-Aug-2025, 05:45 PM ▾

Users profiles:

Courier Management System

Search Shipments, Customers, Tracking No...

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Admin Manager ▾

SETUP

User Profiles

Help for this Page

Profiles

New Profile Edit Delete Create New View

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All

Action	Profile Name ↑	User License	Custom
☐ Edit Clone	Courier Admin	Courier Management User	☒
☐ Edit Clone	Shipment Handler	Courier Management User	☐
☐ Edit Clone	Customer Support	Customer Community User	☐
☐ Edit Clone	Operations Manager	Courier Management User	☒
☐ Edit Clone	Branch Manager	Courier Management User	☐
☐ Edit Clone	Report Viewer	Analytics Cloud Integration User	☐

Conclusion:

The Courier Management System project was successfully built on the Salesforce platform. Custom objects were created to manage customers, shipments, delivery agents, branches, and invoices.

Flows and validation rules were used to automate shipment processing, and reports and dashboards helped track delivery performance and revenue.

This project gave hands-on experience in the complete Salesforce development process — from data modelling to automation, security, and testing. Overall, the CMS application makes courier operations faster, secure, and fully automated.

COURIER MANAGEMENT SYSTEM (CMS)

LEADER: H. GIRIDHARAN

MEMBERS: K. KANNAN

S. MANIVANNAN

M. GOKUL

S.RAGHUL SAJITH

COLLEGE: DHARMAPURAM ADHINAM ARTS

COLLEGE

UNIVERSITY NAME: ANNAMALAI UNIVERSITY

COLLEGE CODE: ANM406